Stata Introductory Workshop

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Ph.D. Public Policy

Stata Introductory Workshop
Level I
2024 Winter Break

KDI SCHOOL
KDI School of Public Policy and Management

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Course Structure

- Modules:
  - Module 1: Stata Fundamentals
  - Module 2: Data Exploration and Graphs
  - Module 3: Data Analysis and Regression
  - Module 4: Applied Research and Replication of Paper
- Schedule:
  - Module 1: Jan 9 Tue: 13:00-16:00
  - Module 2: Jan 11 Thu: 13:00-16:00
  - Module 3: Jan 16 Tue: 13:00-16:00
  - Module 4: Jan 18 Thu: 13:00-16:00

Completion Requirements

- Grade: 'P' on Transcripts (No Credits)
- Assignments[2]:
  - Assignment 1: Deadline: Jan 11 Thu, 1 pm
  - Assignment 2: Deadline: Jan 16 Tue, 1 pm
  - Assignment 3: Deadline: Jan 18 Thu, 1 pm
  - Assignment 4: Deadline: Jan 21 Sun, 11:59 pm
- Others:
  - Attend all offline lectures on specified dates
  - Note: Sign attendance list for acknowledgment

[2] We'll go over the details for the first three assignments at the beginning of the next module.

Outline

1. Module I
2. Way Forward

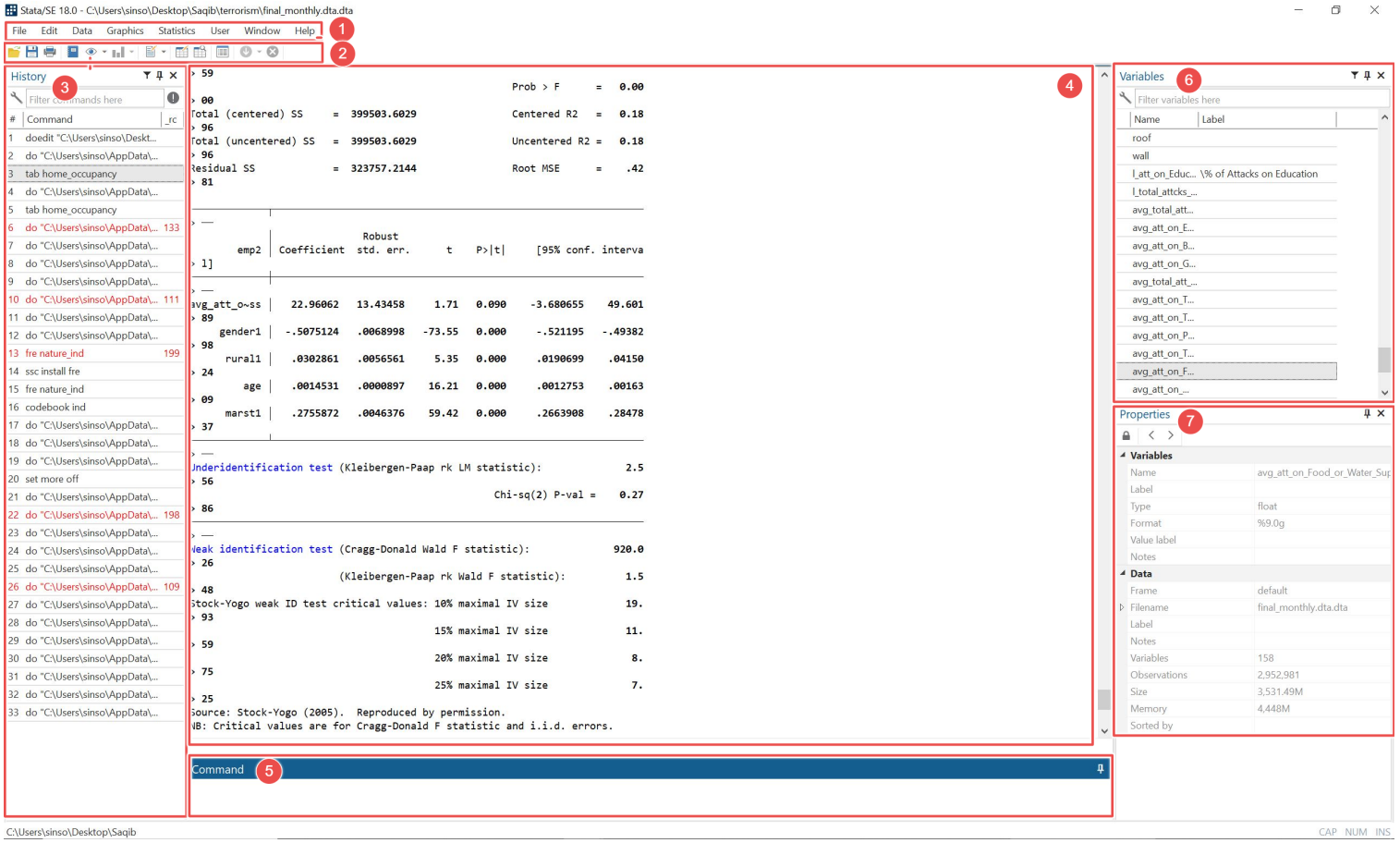
Module I: Stata Fundamentals
Introduction to Stata and Basic Commands

Resources

- Installation Link: `https://www.ekdis.ac.kr/mod/page/view.php?id=41638`
- UCLA Data Analysis: `https://stats.oarc.ucla.edu/`
- Why Stata?: `https://youtu.be/uIo-SInPTqQ`

STATA Interface

1. Menu Bar
2. Quick Buttons
3. History
4. Results
5. Command
6. Variables
7. Properties



Stata Do Files

What is a Stata Do File?

- A Stata `.do` file is a script containing a series of Stata commands.
- It allows you to automate and reproduce your data analysis steps.
- Enhances the reproducibility of your work by documenting and saving commands.

Advantages of Using Do Files

- **Reproducibility:** Easily rerun analyses with the same commands.
- **Documentation:** Provides a clear record of data processing and analysis steps.
- **Efficiency:** Automate repetitive tasks and save time.

Creating a Do File

- Open the Do File Editor in Stata.
- Type or paste your Stata commands into the editor.
- Save the file with a `.do` extension.

Importing and Exporting Data in Stata

Key Commands:

- `use` — Load Stata dataset
- `save` — Save Stata dataset
- `import excel` — Import Excel dataset
- `import delimited` — Import delimited data (csv)

Important Points:

- Stata data files are stored as `.dta` files.
- Data can be loaded from the Stata website or program using `webuse` and `sysuse` commands.
- Creating directories, clearing memory, and managing files using `mkdir`, `clear`, and `dir`.
- Exporting data to Excel (`.xls`) and delimited files, and understanding the differences.
- Saving variable names to the first row during export.
- Importing data from delimited files and Excel, with and without treating the first row as variable names.

Exercise I

- Download a CSV file (`assignment1.csv`) from ekdis and import it
- Save STATA data (`assignment1.dta`)
- Export Excel data (`assignment1.xls`) with variable names to the first row
- Clear
- Import Excel data (`assignment1.xls`)

Viewing Data in Stata

Help Command:

- `help COMMAND` — Open help page for `COMMAND`
- Example: `help summarize` Abbreviations: `summ`, `sum`, `su`

Viewing Data:

- `describe`, `browse`, `list` — Overview commands
- Example:

```
list , li , l
l read write in 1/5, clean
l read write in 1/10, sep(0)
```

Selecting by Condition with `if`:

```
l gender ses math if math > 70
l gender ses read if gender == 2 & read > 70
l gender ses math if math > 70 & ses < 3
```

Exercise II

- List gender, read, and math if math < 40 or read < 40
- List gender, read, and math if females (gender $= 2$) who have math < 40 or read < 40
- List gender, read, math, and prgtype if males (gender $= 1$) not in academic program (prgtype $\neq$ academic) who have read ≥ 55 and math ≥ 45

Exploring Data in Stata

- `codebook` — Inspect variable values
- `summarize` — Summarize distribution
- `tabulate` — Tabulate frequencies

Examples:
Inspect Values of Variables:

```
codebook
codebook read gender prgtype
```

Summarize Variables:

```
sum read math
sum read math if gender == 2
sum read math if gender == 2, detail
```

Tabulate Frequencies of Variables

```
tabulate ses
tabulate ses, nolabel
```

Exercise III

- Tabulate race in descending order of frequency
- Tabulate race and determine the numeric code for "asian" and the numeric code for "white"
- Summarize read, write, and math for Asian
- Summarize read, write, and math for White
- Summarize math by race using the tabulate command
- Tabulate ses and prgtype for White
- Tabulate ses and prgtype for White females who have math ¿ 50
- Summarize read by prgtype for White

Data Management in Stata

- `generate` — Create variable
- `replace` — Replace values of variable
- `egen` — Extended variable generation
- `rename` — Rename variable
- `recode` — Recode variable values
- `label variable` — Give variable description
- `label define` — Generate value label set
- `label value` — Apply value labels to variable
- `encode` — Convert string variable to numeric
- `decode` — Convert numeric variable to string

Examples:

```
gen total = math + science + socst
replace total = math + socst if missing(science)
egen total0 = rowtotal(math science socst)
gen mean = (math + science + socst) / 3
egen mean0 = rowmean(math science socst)
```

Exercise IV

- Generate a variable `ftotal` to sum (all non-missing) 4 variables: `read`, `write`, `math`, `science`
- Replace `ftotal` with a sum of 3 variables (`read + write + math`) if `science` is missing
- Generate a variable `ftotal0` to sum 4 variables: `read`, `write`, `math`, `science`
- Summarize `ftotal` and `ftotal0`
- Generate a variable `fmean` to average (all non-missing) 4 variables: `read`, `write`, `math`, `science`
- Replace `fmean` with a mean of 3 variables (`read` + `write` + `math`)/3 if `science` is missing
- Generate a variable `fmean0` to average 4 variables: `read`, `write`, `math`, `science`
- Summarize `fmean` and `fmean0`

Stata Data Management Commands

Renaming Variables:

```
rename gender female
rename age years
```

Recoding:
Recode variable income: 1 to 0, 2 to 1

```
recode income (1=0)(2=1)
```

Recode variable education: 1 to 0, 2 to 1

```
recode education (1=0)(2=1)
```

Labeling Variables:

```
label variable weight "Weight in kg"
```

Viewing and Tabulating:
List value labels for education

```
label list education
```

Exercise V

- Tabulate `female`
- Label `female` as "female"
- Create a label `female1` (0 "male" 1 "female")
- Apply the label `female1` for `female`
- Tabulate `female`
- Tabulate `race`
- Codebook `race`
- Create a label `racelab` 1 "hispanic" 2 "asian" 3 "black" 4 "white" 5 "etc"
- Apply the label `racelab` for `race`
- Tabulate `race`

Encoding and Decoding in Stata

Encoding a String Variable:
Encode string prgtype into numeric variable prog

```
encode prgtype, gen(prog)
```

Codebook for prgtype and prog

```
codebook prgtype prog
```

Decoding a Numeric Variable:
Decode numeric race into string variable racestr

```
decode race, gen(racestr)
```

Codebook for race and racestr

```
codebook race*
```

Exercise VI

- Create a variable `highmath` that takes 1 if `math > 60` and 0 otherwise using `generate` and `replace` commands
- Tabulate `highmath`
- Summarize `math` by `highmath`
- Label `highmath` as "high math score (`> 60`)"
- Create a label `mathlab` 0 "low" 1 "high"
- Apply `mathlab` for `highmath`
- Tabulate `highmath`
- Create a variable `white` that takes 1 if `racestr = white` and 0 otherwise
- Tabulate `race` and `white`
- Summarize `science` in detail and determine a median
- Create a variable `highscience` that takes 1 if above the median and 0 otherwise
- Tabulate `highscience`

Dataset Operations & Logging in Stata

Operations:

```
keep gender id race ses schtyp prgtype read math
drop schtyp prgtype
keep if gender == 2
keep if math > 50
sort id
gsort -id
order id, first
```

Logging:

```
log using test.txt, text replace
sum science, detail
tab ses
log close
```

Exercise VII

- Open a log file `test.txt`
- Load data `hs0`
- Summarize `math` in detail and determine a median
- Keep if `math` is at the median or above
- Save data as `hs0_highmath`
- Load data `hs0`
- Keep if `math` is below the median
- Save data as `hs0_lowmath`
- Load data `hs0`
- Keep variables: `id` and `write`
- Save data as `hs0_write`
- Close the log file

Thank You for Your Attention!

Recommendations

- Data from Google Drive
- Stata Schemepack
- World Bank Stata Visual Library
- Stata to LaTeX
- Stata Cheatsheet